The phosphosite plot

How many phosphoproteins, phosphopeptides and phosphosites survive a given localisation cutoff.

AT A GLANCE

Where	Phospho → Phosphosite Plot
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded, so filtering and imputation performed elsewhere do not change these counts.
Needs	A phospho dataset. PTM_localization, Protein_group, UPD_seq and PTM_Collapse_key are the columns it reads.
Three bars	Phosphoproteins, phosphopeptides, phosphosites, drawn in descending order with the count printed at the end of each bar.
Shared control	**Localization probability cutoff**, the minimum PTM localisation probability. Everything else is size, format and appearance.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The plot answers how deep the phosphoproteome is, and how much of that depth survives a demand for confident site localisation. Raising the shared localisation cutoff shows how many reported identities meet a stricter requirement. Counts alone do not validate the calibration of the search software's localisation probabilities.

The three numbers are not three measurements of the same thing. They sit in a hierarchy — a protein carries several peptides, a peptide can carry several sites. Their ratios describe the reported identifications, not the fraction of a protein's molecules carrying a phosphate.

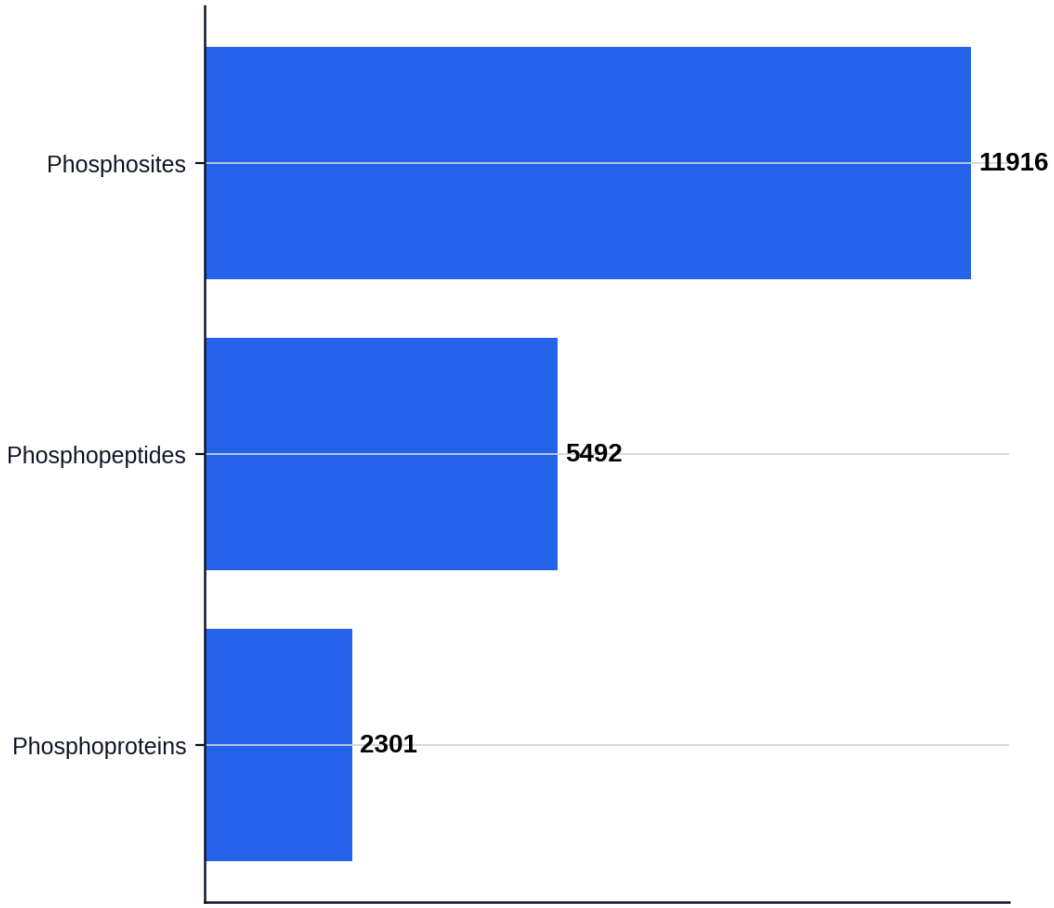


Figure 1.1. The phosphosite plot at **Cutoff** 0 on the example dataset: 11 916 phosphosites, 5 492 phosphopeptides and 2 301 phosphoproteins, drawn in descending order with the count printed at the end of each bar and no ticks on the x axis. Fold change and conditions play no part here — this page counts identifications, not differences.

Reading it

- Phosphoproteins — distinct values of Protein_group among the rows that pass the cutoff. A protein group counts once however many of its sites survive.
- Phosphopeptides — distinct peptide sequences, taken from UPD_seq with every non-word character removed and the rest upper-cased. The modification marker is one of the characters removed, so s*LSPILPGR and SLSPILPGR are the same peptide here.
- Phosphosites — distinct values of PTM_Collapse_key, the site identifier the search output carries.
- The number at the end of each bar. It is always printed; there is no switch for it. The x axis carries no ticks, because the label is the number.

EASY TO GET WRONG

The shared cutoff selects which sites enter phosphosite analyses. Class I is a separate, fixed classification at 0.75. Setting the shared cutoff to 0.75 excludes lower-probability and unknown sites from other phosphosite views too; the coverage plot's remaining sites will therefore be Class I. It does not change the Class I definition.

2 Controls

Controls on this page

Control	What it does	Default	Where
Localization probability cutoff	Shared across phosphosite analyses, between 0 and 1. At zero all sites are retained; above zero a usable probability at least this large is required.	0	module
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	15 × 10, 100	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	The colour of all three bars. Until one is chosen the plot draws its own sky blue, #87CEEB; the picker itself opens on #2563eb.	#87CEEB	appearance
Title, axis labels	Text, size, weight, colour and position of every label, and whether the title is drawn at all. This plot sets no title and no axis labels of its own, so the fields start empty.	none	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid	Which spines are drawn, and the grid over them. The top and right spines are off and the grid is off before the panel touches them. Tick styling is not offered here: the plot clears its x ticks, so there is nothing to style.	left and bottom spines, no grid	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

Above a shared cutoff of zero, PTM_localization is read as a number and only rows meeting the cutoff are kept. At zero the selection is disabled. Over the selected rows three counts are taken:

- phosphoproteins: distinct non-missing Protein_group;
- phosphopeptides: distinct non-empty sequences after UPD_seq has had every character outside [A-Za-z0-9_] removed and been upper-cased;
- phosphosites: distinct non-missing PTM_Collapse_key.

The three are sorted by count, largest first, and drawn as horizontal bars with the value written at the end of each. Nothing is divided by anything and no sample column is read: a site counts once whether it was quantified in one run or in all of them.

Edge cases

- At cutoff 0, missing probabilities do not remove rows. Above 0, they cannot meet the requirement and are excluded. A positive cutoff with no localisation column is refused; set it to 0 or supply a suitable probability column.
- If the cutoff removes every row the plot is replaced by the words No phospho rows available and the table comes back empty.
- Without UPD_seq the phosphopeptide bar falls back to the number of distinct PTM_Collapse_key values, which makes it equal to the phosphosite bar. Two bars of exactly the

same height are the symptom.

- Sample values, replicates and missing values never enter this page; it counts identifiers.

4 The table below the plot

Columns in this table

Column	What it holds
Term	Phosphoproteins, Phosphopeptides or Phosphosites.
Count	The number behind that bar, at the current **Cutoff**.

The table is sorted the same way as the bars and downloads as CSV.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = simple_phos_site_plot(data3)
```

```r
fig <- simple_phos_site_plot(data3)
```

The function takes the phosphosite frame alone — no annotation, because no sample column is read. **Cutoff** is the filter_value argument and defaults to 0, and color, width_cm, height_cm and dpi stand in for the appearance and size controls of the page; unlike the page, the function has no fallback for a frame without UPD_seq and will fail on one rather than repeating the phosphosite count.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

In well-populated datasets the phosphopeptide and phosphosite bars stand well above the phosphoprotein bar, which is the hierarchy showing through. Counts that barely move as **Cutoff** is raised from 0 to 0.75 indicate confidently localised identifications; a sharp fall over the same range means a large share of the sites are assigned to a residue the evidence does not fix, and any site-level conclusion drawn from that dataset should be checked against the Class I count on the coverage page first.

Behind this page

Function	What it produces
simple_phos_site_plot	The bar chart, in the Python package
phosphosite_plot_table	The two-column table of counts
phosphosite_plot_png	The figure the page draws